

Spike Firefly

The **L-Spike 1x**, formerly the **Spike Firefly**, is a loitering munition built by Rafael Advanced Defense Systems.

Design

The Spike Firefly was developed to provide infantry forces with the capability to engage enemies behind cover during urban warfare; although it isn't an anti-tank guided missile (ATGM) like the rest of the Spike missile family, it is considered part of it from using components developed for them. A battery powers two electric motors that drive two counter-rotating two-blade rotors allowing for vertical takeoff and landing (VTOL) and hovering with minimal noise. The munition body consists of three sections: a sensor package that includes an uncooled thermal imager and EO day sensor with a proximity sensor that can track and pursue agile targets; the battery that delivers endurance of 15 minutes; and a payload section. The payload can be a 350 g omnidirectional blast fragmentation warhead in a strike mode or another battery doubling endurance to 30 minutes in a reconnaissance mode. It can operate autonomously flying via waypoints, or be controlled by an operator to search and attack targets beyond line of sight. The operator-in-the-loop function allows for a strike to be aborted if needed, and it can return to be reused.^{[1][2]}

The Spike Firefly body section is 400 mm tall, 80 mm wide, and weighs 3 kg. Control range is 1,500 m in open terrain and 500 m in an urban setting. It can fly at 60 km/h (17 m/s), reach 70 km/h (19 m/s) when diving for an attack, and can operate in wind speeds up to 36 km/h. A Spike Firefly system contains three munitions held in tubes and a 1 kg

L-Spike 1x



Type	<u>Loitering munition</u>
Place of origin	 <u>Israel</u>
Service history	
In service	2020-present
Used by	<u>Israeli Defense Forces</u>
Wars	<u>Gaza War</u>

Production history

Designed	2018
Manufacturer	<u>Rafael Advanced Defense Systems</u>
Produced	2018–present

Specifications

Mass	2.2 kg
Width	80 mm
Height	400 mm

Warhead	420 g
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Engine	<u>Electric motor</u>
Operational range	5,000 m
Maximum speed	60 km/h (maneuvering) 70 km/h (diving)

control unit carried in a backpack by a single operator totaling 15 kg.^{[1][2]} It is positioned to be operated at company-level. The upgraded L-Spike 1x weighs 2.2 kg, has a 420 g warhead, and has operational range extended to 5,000 m.^{[3][4]}

Operational history

Rafael unveiled the Spike Firefly in June 2018.^{[1][5]} In May 2020, the Israel Defense Forces (IDF) announced they were procuring the munition and calling it **Maoz** in IDF service.^{[6][7]} It was first used in combat during the July 2023 Jenin incursion.^{[8][9]} Following the October 7 attacks, the Spike Firefly was used in operations in the Gaza Strip during the Gaza War.^[10] In 2025, Rafael renamed an upgraded version of the Spike Firefly to the L-Spike 1x.^[3]

Operators

-  Israel
-  Vietnam: Manufactured domestically^[11]

See also

- STM Kargu
- IAI Rotem L
- Uvision Hero-R
- Edge Group QX-1

References

1. Rafael: a new member for the Spike family (<https://www.edrmagazine.eu/rafael-a-new-member-for-the-spike-family>). *European Defence Review*. 18 June 2018.
2. Israel's FireFly Urban Warfare Drone Spotted In Jenin (<https://www.twz.com/israels-firefly-urban-warfare-drone-spotted-in-jenin>). *The War Zone*. 18 December 2023.
3. Rafael's new backpack-sized missile gives ground forces precision strike capability (<https://www.jpost.com/defense-and-tech/article-874601>). *The Jerusalem Post*. 20 November 2025.
4. Rafael unveils new SPIKE missile (<https://en.globes.co.il/en/article-rafael-unveils-new-spike-missile-1001527001>). *Globes*. 20 November 2025.
5. FireFly: New variant in Rafael's Spike family of precision weapons (<https://www.jpost.com/israel-news/firefly-new-variant-in-rafaels-spike-family-of-precision-weapons-559228>). *The Jerusalem Post*. 5 June 2018.
6. Israeli MOD orders RAFAEL's Spike FireFly loitering munition for IDF ground forces (<https://www.edrmagazine.eu/israeli-mod-orders-rafaels-spike-firefly-loitering-munition-for-idf-ground-forces>). *European Defence Review*. 4 May 2020.
7. Israel acquires FireFly loitering munition for close combat (<https://www.c4isrnet.com/unmanned/2020/05/05/israel-acquires-firefly-loitering-munition-for-close-combat/>). *C4ISRNET*. 5 May 2020.

8. After use in combat, Rafael looks to future for Spike FireFly loitering munition (<https://breakingdefense.com/2023/07/after-use-in-combat-rafael-looks-to-future-for-spike-firefly-loitering-munition/>). *Breaking Defense*. 14 July 2023.
9. Rafael confirms use of Firefly compact loitering munitions (<https://en.globes.co.il/en/article-rafael-confirms-deployment-of-firefly-miniature-loitering-munitions-1001483738>). *Globes*. 8 July 2024.
10. Israel uses new defense technology on Gaza battlefield (https://www.fdd.org/analysis/op_ed/s/2023/12/19/israel-uses-new-defense-technology-on-gaza-battlefield/). *Foundation for Defense of Democracies*. 19 December 2023.
11. Sato, Daisuke. "Vietnam signs deal for Israel-made Firefly loitering munition" (<https://defence-blog.com/vietnam-signs-deal-for-israel-made-firefly-loitering-munition/>). Defence Blog. Retrieved 5 March 2026.

External links

- SPIKE® FIREFLY Miniature Loitering Munition (<https://www.rafael.co.il/wp-content/uploads/2024/08/spike-firefly-miniature-loitering-munition.pdf>)
 - L-SPIKE® 1X Tactical Loitering Munition - Rafael (<https://www.rafael.co.il/system/l-spike-1x/>)
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